

React Fundamentals

The Power of React



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No React knowledge needed



**[https://github.com/RolandGuijt/
react-fundamentals](https://github.com/RolandGuijt/react-fundamentals)**



<https://code.visualstudio.com/download>



What to Expect?

Everything that's needed

Prerequisite: javascript

Sit back/code along

GitHub repository

Visual Studio Code



React

**Open-source javascript library for
building user interfaces**



Where is React Used?

Single Page Applications (SPAs)

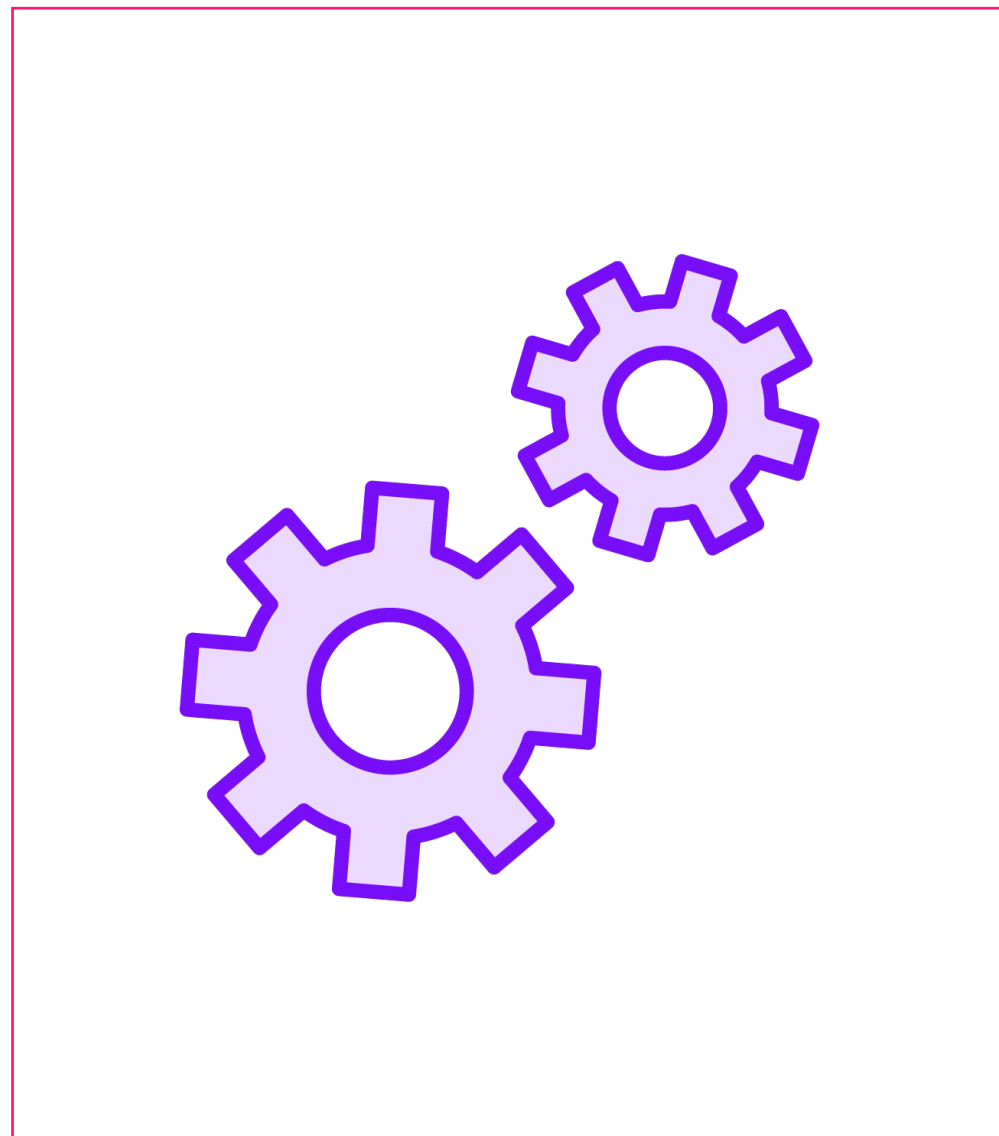
Mobile applications (React Native)

Principles the same throughout

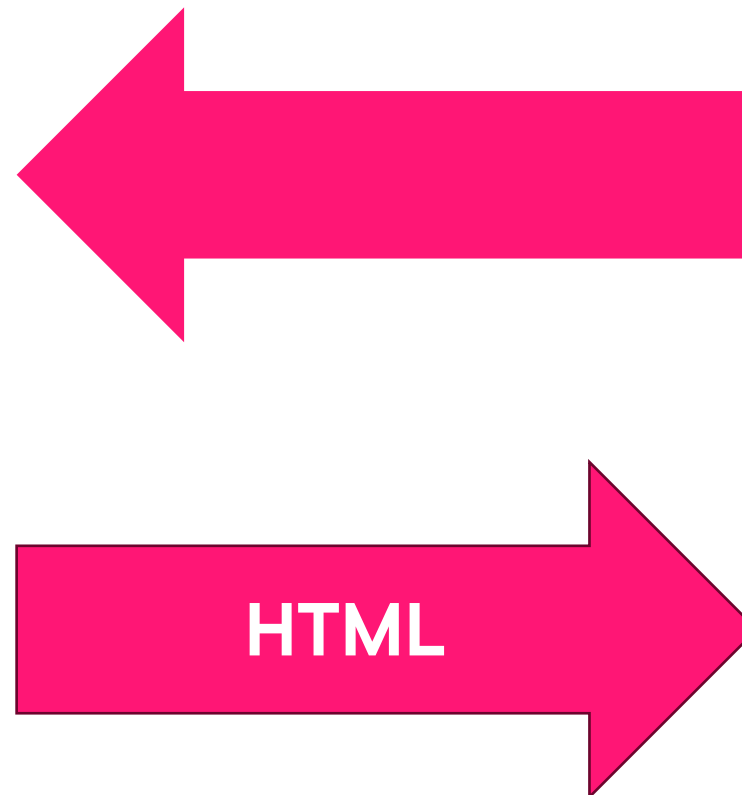


Traditional Web Application

Server



Browser

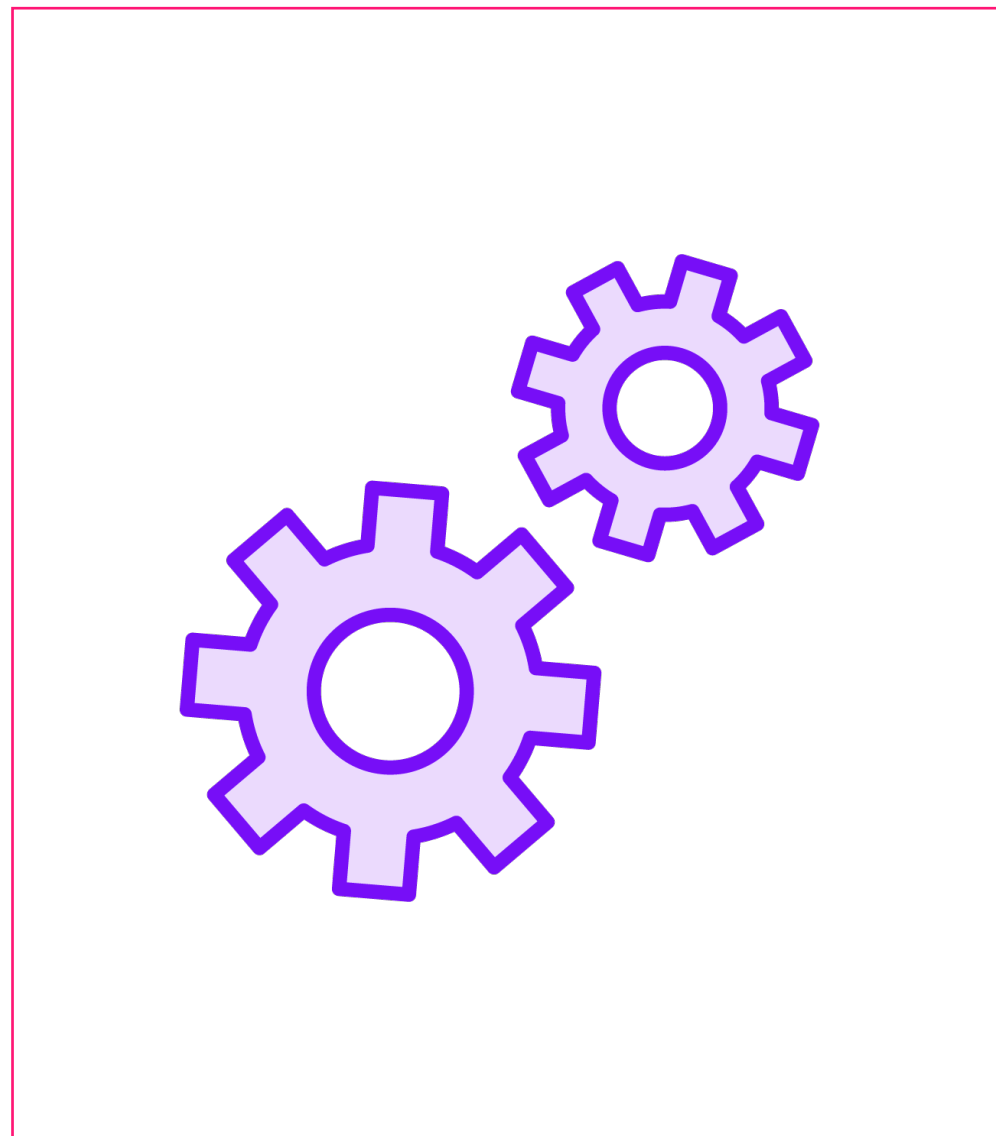


Submit

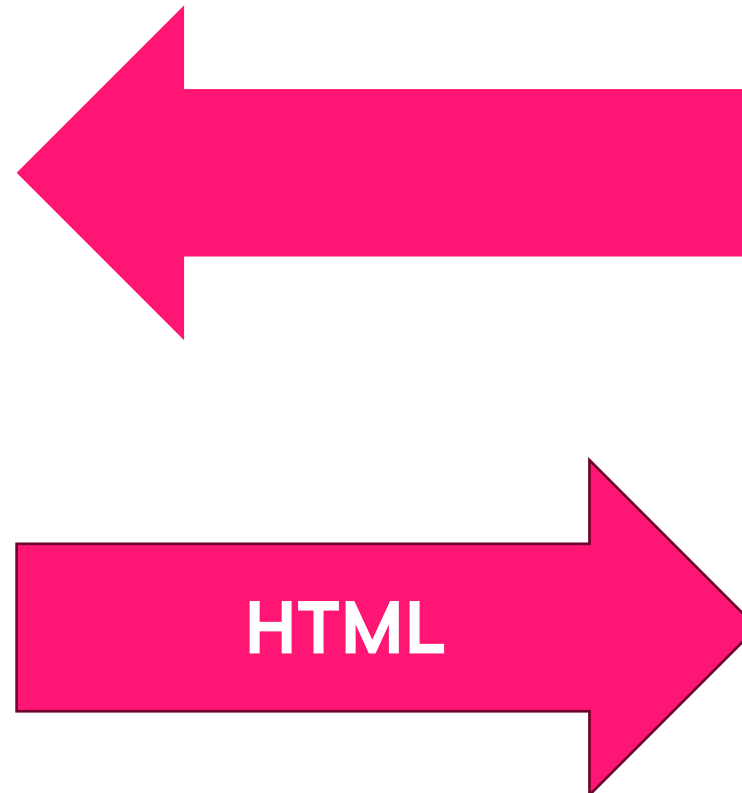


Traditional Web Application

Server



Browser

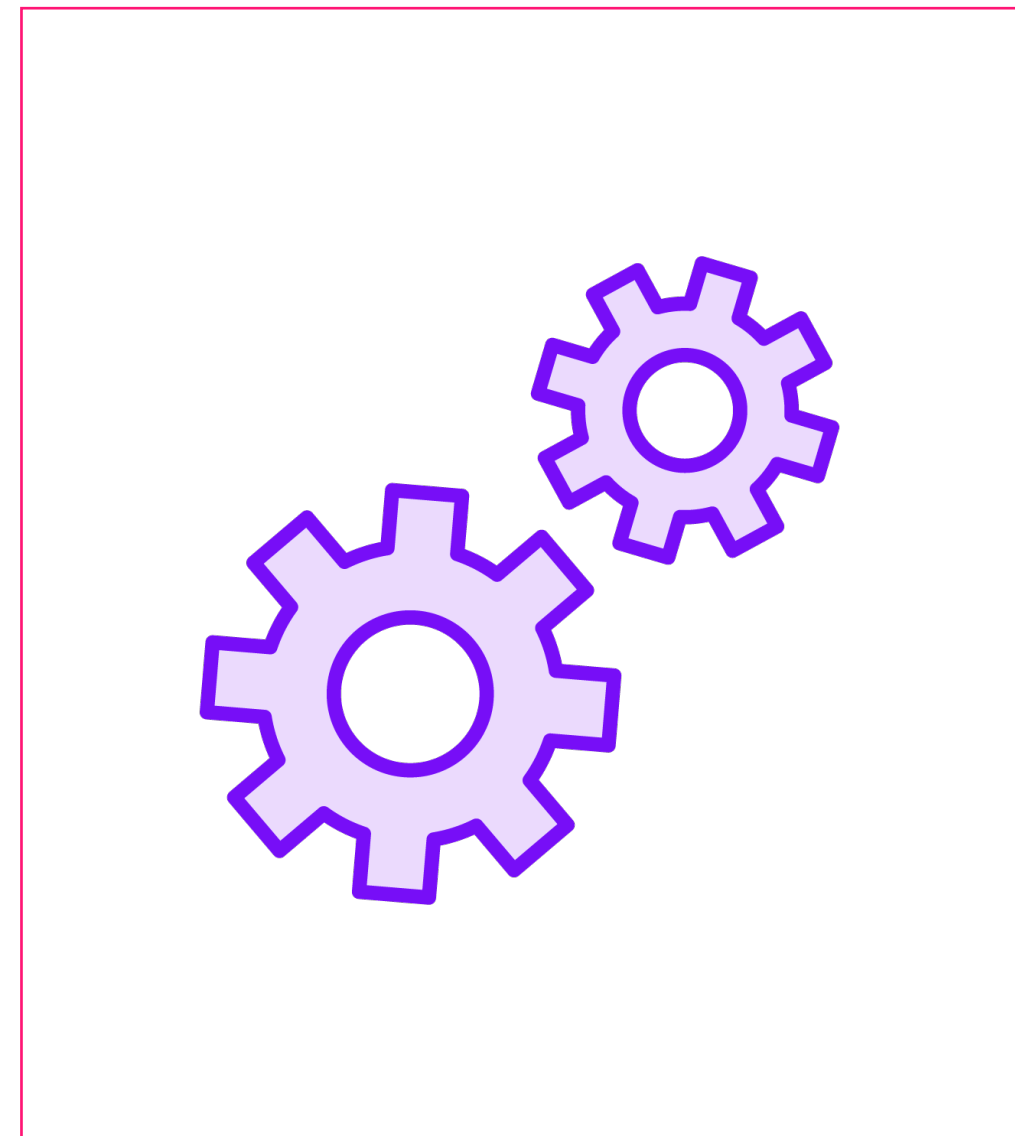


Single Page Application (SPA)

Server



Browser

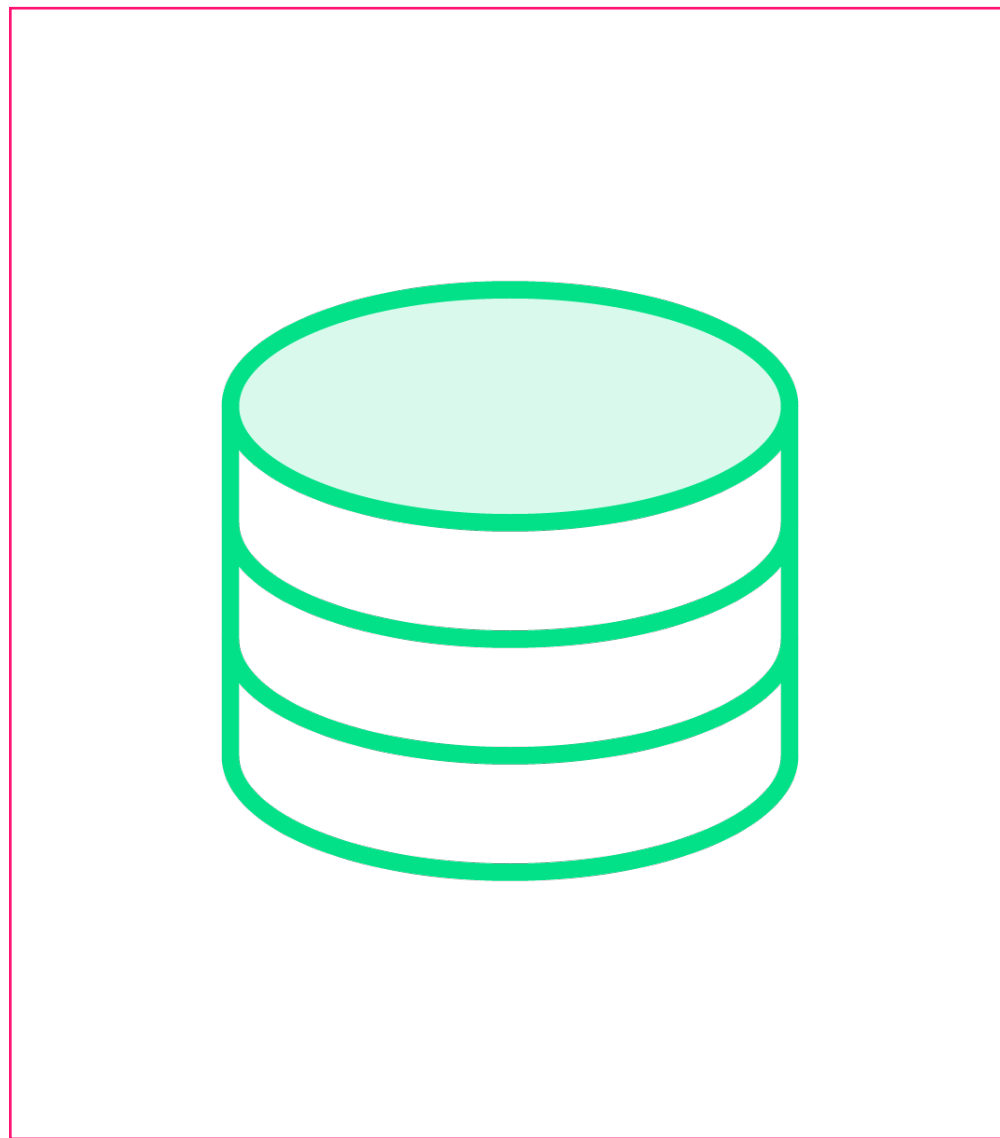


1st request

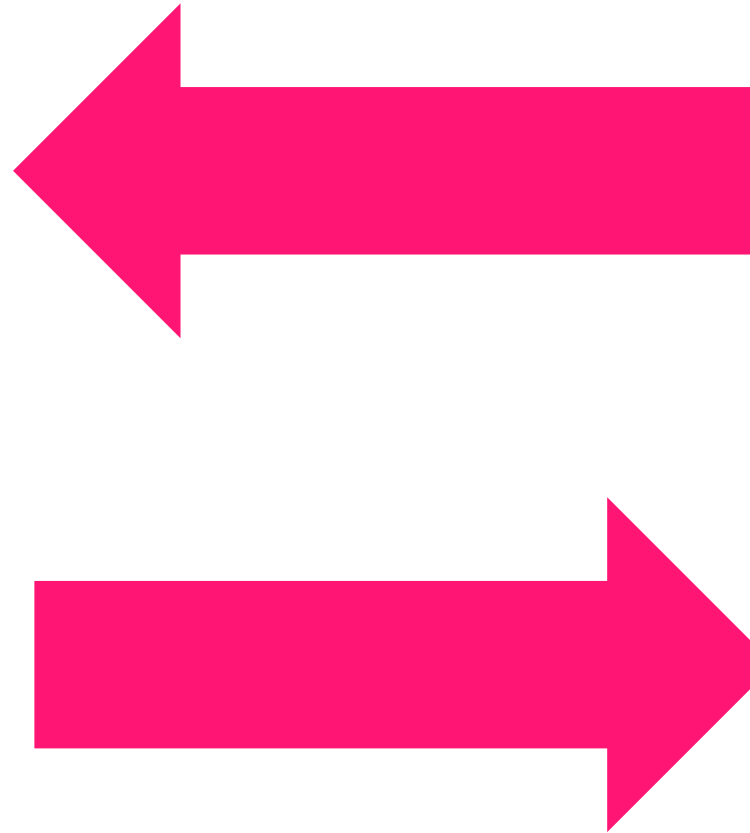
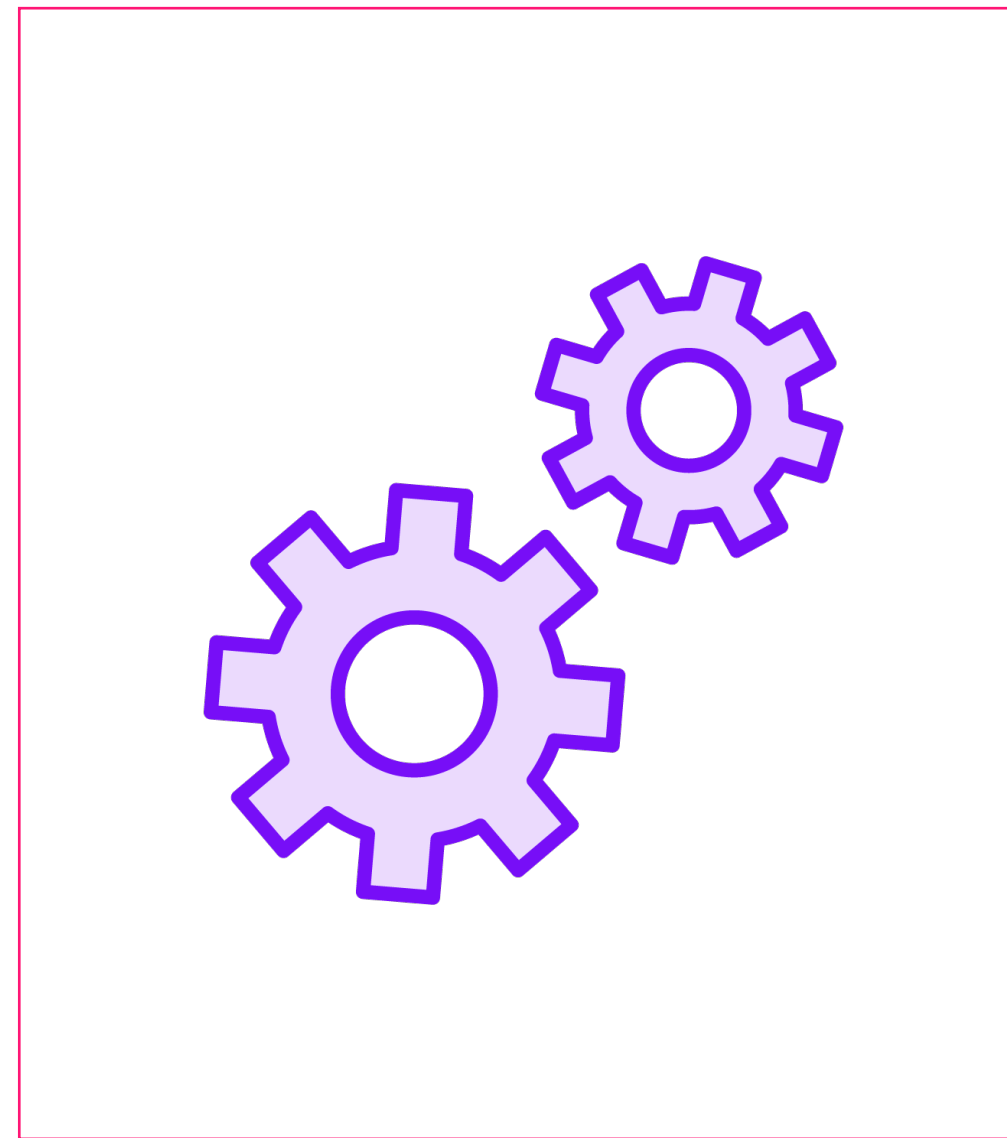


Single Page Application (SPA)

API



Browser



React

JavaScript library for building user interfaces

Enables structured, maintainable logic

Browser-side or server-side

Focusses on just the user interface



Reconciliation

New DOM is compared to old one

Difference is applied in browser

With more complex UIs this also works

Result: faster and better user experience





Components

JavaScript function

Exists in separate file/module

Maintains state

Function returns UI using JSX

When state changes function re-executes

UI is updated using reconciliation



Working with Components

Reusability

Modularity

**Separation of
Concerns**

**Improved Readability
and Maintainability**

Easier Testing



A Component

```
const Banner = () => <h1>This is a banner</h1>;
```



JSX

Components are JavaScript functions that return JSX

JSX looks like HTML but it isn't

An alternative way to write JavaScript

JSX: JavaScript eXtension

Translated to JavaScript by a tool

Babel



Separation of Concerns

react

react-dom



JSX to Markup

```
<h1>  
  This is a banner  
</h1>
```

Transpiler

```
React.createElement(  
  "h1", null, "This is a banner");
```

React DOM

```
<h1>  
  This is a banner  
</h1>
```



Advantages

Reconciliation

JavaScript can be used in the UI



Another Component

```
const Greeting = () => (  
  <div>  
    <Banner />  
    <h2 className="highlight">Greetings!</h2>  
  </div>  
) ;
```



Built-in Components

Correspond to DOM elements

Different output targets use different built-in components

Are camelCased

Custom components are PascalCased



Next up:
Setting Up a New Project

